EX.NO.1	Create a responsive web page with a registration form using HTMLS, CSS3, and Bootstrap grid system.

AIM:

To design a player registration form using HTML, CSS, and Bootstrap.

DESCRIPTION:

➢ The <head> section contains metadata, Bootstrap link, and internal CSS.
➢ The <meta charset="UTF-8"> sets the character encoding.
➢ The <title> tag sets the title of the webpage.
➢ The <link> tag is used to include Bootstrap for styling.
➢ The <style> tag is used to apply internal CSS.
➢ The body selector sets the background color of the page.
➢ The .form-box class styles the form container with background, padding, border radius, and shadow.
➢ The .header class styles the header with background color, spacing, alignment, and text color.
➢ The .header img styles the images with height and rounded corners.
➢ The <body> contains all visible content.
➢ The creates a Bootstrap container for layout.
➢ The acts as the main form container.
➢ The contains the title and images.
➢ The <img> tags display images in the header.
➢ The <h3> tag displays the heading "IPL Player Registration".
➢ The <form> element is used to collect user input.
➢ The row class is used to create horizontal groups of columns.
➢ The col-md-* classes divide the row into responsive columns.
➢ The <label> tag is used to describe input fields.
➢ The <input> tag is used to create input fields.
➢ The type="text" creates a text input field.
➢ The type="number" creates a number input field.
➢ The type="date" creates a date picker.
➢ The type="email" creates an email input field.
➢ The <select> tag is used to create dropdown lists.
➢ The <option> tag defines items in the dropdown.
➢ The <textarea> tag is used for multi-line text input.
➢ The <button> tag creates a submit button.
➢ The btn and btn-success classes style the button using Bootstrap.
➢ The text-center class centers the button.

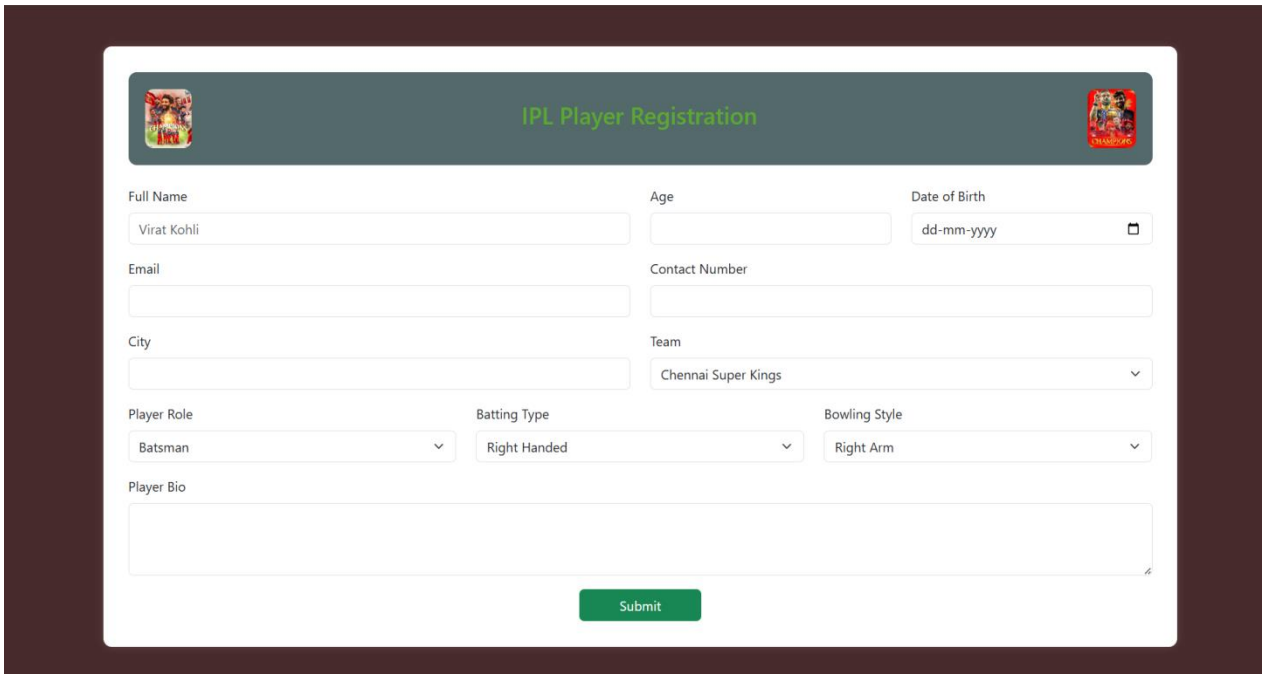
SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>IPL Player Registration</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
<style>
body{
background-color:#462b2b;
}
.form-box{
background:white;
padding:30px;
border-radius:10px;
box-shadow:0 0 10px rgb(211, 194, 194, 0.2);
}
.header{
background:#536a6a;
padding:20px;
border-radius:10px 10px ;
color:rgb(93, 168, 49);
display:flex;
justify-content:space-between;
align-items:center;
}
.header img{
height:70px;
border-radius:10px;
}
</style>
</head>
<body>
<div class="container mt-5">
<div class="form-box">
<div class="header">
<img src="C:/Users/pravi/Downloads/images (5).jpg">
<h3>IPL Player Registration</h3>
<img src="C:/Users/pravi/Downloads/images (6).jpg">
</div>
```

```
<form class="mt-4">
<div class="row mb-3">
<div class="col-md-6">
<label class="form-label">Full Name</label>
<input type="text" class="form-control" placeholder="Virat Kohli">
</div>
<div class="col-md-3">
<label class="form-label">Age</label>
<input type="number" class="form-control">
</div>
<div class="col-md-3">
<label class="form-label">Date of Birth</label>
<input type="date" class="form-control">
</div>
</div>
<div class="row mb-3">
<div class="col-md-6">
<label class="form-label">Email</label>
<input type="email" class="form-control">
</div>
<div class="col-md-6">
<label class="form-label">Contact Number</label>
<input type="text" class="form-control">
</div>
</div>

<div class="row mb-3">
<div class="col-md-6">
<label class="form-label">City</label>
<input type="text" class="form-control">
</div>
<div class="col-md-6">
<label class="form-label">Team</label>
<select class="form-select">

<option>Chennai Super Kings</option>
<option>Mumbai Indians</option>
<option>Royal Challengers Bangalore</option>
<option>Kolkata Knight Riders</option>
</select>
</div>
</div>
<div class="row mb-3">
<div class="col-md-4">
```

```
<div class="col-md-4">
<label class="form-label">Bowling Style</label>
<select class="form-select">
<option>Right Arm</option>
<option>Left Arm</option>
</select>
</div>
</div>
<div class="mb-3">
<label class="form-label">Player Bio</label>
<textarea class="form-control" rows="3"></textarea>
</div>
<div class="text-center">
<button class="btn btn-success px-5">Submit</button>
</div>
</form>
</div>
</div>
</body>
</html
```

OUTPUT:



RESULT:

A responsive IPL Player Registration form is designed and implemented using HTML, CSS, and Bootstrap.

EX.NO.2	**Enhance web elements with CSS: Shadows, Rounded Borders, and Multiple Backgrounds.**

AIM:

To create a personal profile webpage using basic HTML and CSS styling.

DESCRIPTION:

- The <html> tag is the root element of the webpage.
- The <head> section contains styles for the webpage.
- The <style> tag is used to apply internal CSS.
- The #r selector is used to style the main container.
- The border-style: double creates a double border around the container.
- The border-color sets the color of the border.
- The border-radius gives rounded corners to the container.
- The box-shadow adds a shadow effect around the box.
- The text-align: center centers all content inside the container.
- The text-shadow adds a shadow effect to the text.
- The font-family changes the style of the text.
- The margin-left and margin-right create space on both sides.
- The img selector is used to style the image.
- The border-radius in img makes the image corners rounded.
- The <body> tag contains the visible content of the page.
- The is used as a container for all content.
- The text "V.PRAVINRAJ" is displayed as a heading.
- The <br> tag is used to insert a line break.
- The <img> tag is used to display the profile image.
- The height and width attributes set the image size.
- The <h6> tag is used to display the description text.
- The h6 CSS removes text shadow and font style for the paragraph.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
<head>
<style>
#r{
  border-style:double;
  border-color:rgb(13, 11, 9);
  border-radius: 20px;
  box-shadow: 3px 3px 20px rgb(255, 14, 14);
  text-align:center;
  text-shadow:2px 2px 8px rgb(76, 9, 220);
  font-family:'Gill Sans', 'Gill Sans MT', Calibri, 'Trebuchet MS', sans-serif;
  margin-left:50px;
  margin-right:50px;
}
img{
    border-radius: 20px;
}
h6{
    text-shadow:none;
    font-family:none;
}
</style>
</head>

<body>
<div id="r">
    V.PRAVINRAJ <br>
    <img src="C:/Users/pravi/Downloads/WhatsApp Image 2026-01-09 at 9.07.22 PM.jpeg"
height="95" width="80">
    <h6>
        Hi this is Pravinraj V. I am studying first year at SKCET college Computer Science
engineering.I love travelling around the worldand
        exploring the culture follwed by the people.My hobbie is playing cricket.
    </h6>
</div>
</body>
</html>
```

OUTPUT:

RESULT:

Web elements are enhanced using CSS features like shadows, rounded borders, and multiple backgrounds.

EX.NO.3	Implement a CSS grid-based layout to create a responsive web page.

AIM:

To implement a CSS grid-base layout to create a responsive web page.

DESCRIPTION:

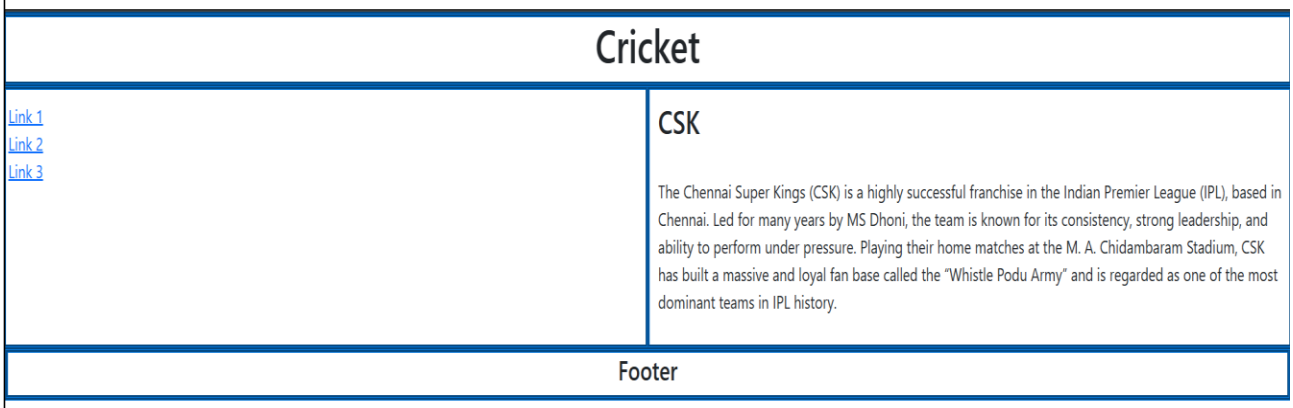
- The <!DOCTYPE html> declares the document type as HTML5.
- The <html> tag is the root element of the webpage.
- The <head> section contains the Bootstrap link and internal CSS.
- The <link> tag is used to include Bootstrap for styling.
- The <style> tag is used to apply internal CSS.
- The div selector applies a groove border and color to all div elements.
- The .row class is styled with display: flex to arrange items in a row.
- The .col class is used to create equal-width columns with padding.
- The <body> tag contains all the visible content of the webpage.
- The acts as the main container for the page.
- The first contains the main heading.
- The tag is used to center the heading text.
- The <h1> tag displays the title "Cricket".
- The second divides content into two columns.
- The first contains navigation links.
- The <a> tags are used to create clickable links.
- The <br> tags are used to insert line breaks between links.
- The second contains text content about CSK.
- The <h3> tag displays the subheading "CSK".
- The <p> tag is used to display the paragraph description.
- The last represents the footer section.
- The <h4> tag displays the footer text "Footer".
- The layout uses flexbox to arrange content side by side.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
<head>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
<style>
div{
border-style: groove;
border-color: rgb(6, 117, 213);
}
.row{
display: flex;
}
.col{
flex : 1;
padding: 10px;
}
</style>
</head>
<body>
<div class="head">
<div class="row">
<center>
<h1>Cricket</h1>
</center>
</div>
<div class="row">
<div class="col">
<a href="#"> Link 1</a><br>
<a href="#"> Link 2</a><br>
<a href="#"> Link 3</a><br><br><br>
</div>
<div class="col">
<h3>CSK</h3><br>
<p>
The Chennai Super Kings (CSK) is a highly successful franchise in the Indian Premier
League (IPL), based in Chennai. Led for many years by MS Dhoni, the team is known for
its consistency, strong leadership, and ability to perform under pressure. Playing their
home matches at the M. A. Chidambaram Stadium,
CSK has built a massive and loyal fan base called the
“Whistle Podu Army” and is regarded as one of the most dominant teams in IPL history.
</p>
```

```
 </div>
</div>
<div class="fooler">
<center>
<h4>Footer</h4>
</center>
</div>
</div>
</body>
</html>
```

OUTPUT:



RESULT:

A responsive web page layout is implemented using CSS Grid.

EX.NO.4	**Implement a JavaScript-based form validation for user input fields (email, password, etc.).**

AIM:

To implement a Javascript-based form validation for user input fields(email, password, etc.).

DESCRIPTION:

➢ The <!DOCTYPE html> defines the document type as HTML5.
➢ The <html> tag is the root element of the webpage.
➢ The <head> section contains style and script information.
➢ The <style> tag is used to apply CSS to the page.
➢ The p selector sets paragraph text color to green.

➢ The <script> tag contains JavaScript code.
➢ The validate() function is used to validate inputs.
➢ document.getElementById() is used to access HTML elements.
➢ The .value property retrieves user input values.

➢ A regular expression checks if the phone number has 10 digits.
➢ Another regular expression validates the email format.
➢ The if-else statement is used to check valid or invalid input.
➢ The inner HTML property displays results in paragraph tags.

➢ The <form> element is used to group input fields.
➢ The <input> and <button> elements are used for user interaction, and <br> is used for spacing.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
<head>
   <STYLE>
      p{
         color: green;
      }
   </STYLE>

   <script>
      function validate(){
         var ph = document.getElementById("phone").value;
         var pattern=/^[0-9]{10}$/;
         if(pattern.test(ph))
            var res="valid phone number";
         else
            res="invalid phone number";

         document.getElementById("para").innerHTML=res;

         var a=document.getElementById("email").value;
         var formate=/^[A-Z a-z 0-9]+@[A-Z a-z]+.(com|ac\.in|co\.in)$/

         if(formate.test(a))
            var tt="email valid";
         else
            tt="invalid email id";

         document.getElementById("p").innerHTML=tt;
      }
   </script>
</head>

<body>
   <form>
      Phone No:
      <input type="text" placeholder="PHONE NUMBER" id="phone">

      <p id="para"></p>

      <input type="button" value="click" onclick="validate()">
```

```
<br>
        <br>

        Email id:
        <input type="text" placeholder="Email          " id="email">

        <p id="p"></p>

        <input type="button" value="click" onclick="validate()">
    </form>
</body>
</html>
```

OUTPUT:

Phone No: PHONE NUMBER

click

Email id: Email

click

RESULT:

JavaScript-based form validation is implemented to verify user inputs like phone number and email.

EX.NO.5	Develop a dynamic webpage using JavaScript and DOM manipulation.

AIM:

To create a simple calculator using HTML, CSS, and JavaScript.

DESCRIPTION:

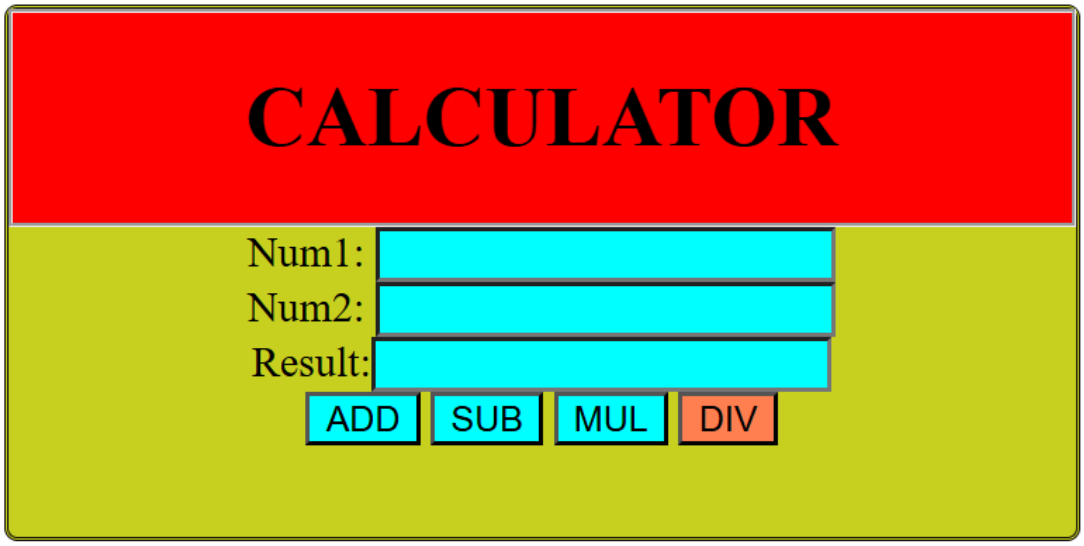
➢ The <!DOCTYPE html> defines the document type as HTML.
➢ The <html> tag is the root element of the webpage.
➢ The <head> section contains internal CSS styling.
➢ The <style> tag is used to apply CSS.
➢ The #r selector styles the main container with border, background color, margin, height, and alignment.
➢ The #g selector styles the header section with background color and border.
➢ The #o selector is used to style the DIV button with a different color.
➢ The input selector applies background color to all input fields.
➢ The <body> contains the visible content of the page.
➢ The acts as the main container.
➢ The contains the calculator heading.
➢ The <h1> tag displays the title "CALCULATOR".
➢ The <input> tags are used to take user input for numbers.
➢ The type="text" creates input fields for numbers and result.
➢ The <br> tag is used to insert line breaks.
➢ The <input type="button"> creates buttons for operations.
➢ The onclick attribute is used to call JavaScript functions.
➢ The <script> tag is used to write JavaScript code.
➢ The getElementById() method is used to access input values.
➢ The parseInt() function converts string input into numbers.
➢ The sum() function performs addition.
➢ The sub() function performs subtraction.
➢ The mul() function performs multiplication.
➢ The div() function performs division.
➢ The result is displayed in the result input field.

SOURCE CODE:

```
!DOCTYPE html>
<html>
<head>

<style>
#r{
border-style:double;
border-color:rgb(13, 11, 9);
background-color:rgb(199, 208, 31);
border-radius: 5px;
margin-left:300px;
margin-right:300px;
text-align:center;
height: 200px;
}
#g{
background-color:red;
border-style:groove;

}

#o{
background-color:coral;
}


input{
background-color:aqua
}
</style>
</head>
<body >
<div id="r">
<div id="g">
<h1>CALCULATOR</h1>
</div>
Num1: <input type="text" id="num1"><br>
Num2: <input type="text" id="num2"><br>
```

```
Result:<input type="text" id="res"><br>
<input type="button" value="ADD" onclick="sum()">
<input type="button" value="SUB" onclick="sub()">
<input type="button" value="MUL" onclick="mul()">
<input type="button" value="DIV" id="o" onclick="div()">

<!-- <p id="res"></p> -->
<script>
function sum()
{
var a=document.getElementById("num1").value;
var b=document.getElementById("num2").value;
document.getElementById("res").value=parseInt(a)+parseInt(b);
}

function sub(){
var a=document.getElementById("num1").value;
var b=document.getElementById("num2").value;
document.getElementById("res").value=parseInt(a)-parseInt(b);
}

function mul(){
var a=document.getElementById("num1").value;
var b=document.getElementById("num2").value;
document.getElementById("res").value=parseInt(a)*parseInt(b);
}
function div(){
var a=document.getElementById("num1").value;
var b=document.getElementById("num2").value;
document.getElementById("res").value=parseInt(a)/parseInt(b);
}
</script>

</div>

</body>
</html>
```

OUTPUT:



\

RESULT:

A dynamic webpage is developed using JavaScript and DOM manipulation to perform calculator operations.

EX.NO.6	Work with APIs and fetch data asynchronously using JavaScript.

AIM:

To work with APIs and fetch data asynchronously using JavaScript.

DESCRIPTION:

1. **Start the program**
   Load the HTML page in a browser.

2. **Display button interaction**
   Wait for the user to click the **DISPLAY** button.

3. **Trigger function**
   When the button is clicked, call the display () function.

4. **Fetch data from API**

   a. Send an asynchronous request to the URL:
      https://jsonplaceholder.typicode.com/users

   b. Wait for the response.

5. **Convert response to JSON**
   Parse the response into a JavaScript object (data).

6. **Initialize output string**
   Create an empty string variable output.

7. **Iterate through user data**
   For each user in the data array:

   a. Extract the name field.

   b. Extract the city field from address.city.

   c. Append both values to the output string with formatting.

8. **Update HTML content**
   Insert the output string into the element with id "result".

9. **End program**
   The names and cities of all users are displayed on the webpage.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
   <body>
      <H1> API INTERFACE PROGRAM</H1>
      <br><br>
      <button onclick="display()">DISPLAY</button>
      <br>
      <p id="result"></p>
      <script>
       async function display()
        {
           let info=await fetch("https://jsonplaceholder.typicode.com/users");
           let data=await info.json();
           let output="";
           for(let i=0;i<data.length;i++)
        {
                output+="<br>Name:"+ data[i].name+ "<br>City:"+ data[i].address.city+"<br>";
        }
           document.getElementById("result").innerHTML=output;
       }
      </script>
   </body>
</html>
```

OUTPUT:

API INTERFACE PROGRAM

DISPLAY

Name:Leanne Graham
City:Gwenborough

Name:Ervin Howell
City:Wisokyburgh

Name:Clementine Bauch
City:McKenziehaven

Name:Patricia Lebsack
City:South Elvis

Name:Chelsey Dietrich
City:Roscoeview

Name:Mrs. Dennis Schulist
City:South Christy

Name:Kurtis Weissnat
City:Howemouth

Name:Nicholas Runolfsdottir V
City:Aliyaview

Name:Glenna Reichert
City:Bartholomebury

Name:Clementina DuBuque
City:Lebsackbury

RESULT:

JavaScript enables smooth interaction with APIs through asynchronous data fetching. Using fetch() and promises, data can be retrieved and processed without blocking the UI. This approach enhances responsiveness and demonstrates real-time integration of external data into web applications.

EX.NO.7	Create a TypeScript application demonstrating object-oriented programming concepts.

AIM:

To demonstrate class inheritance, encapsulation, and method overriding using TypeScript.

DESCRIPTION:

- ➢ The class person is used to define a base class.
- ➢ The public keyword allows access to variables from outside the class.
- ➢ The private keyword restricts access to the variable within the class only.
- ➢ The name, age, and city are attributes of the person class.
- ➢ The constructor is used to initialize the values of attributes.
- ➢ The this keyword refers to the current object.
- ➢ The display() method is used to print the person details.
- ➢ The console.log() is used to display output in the console.
- ➢ The getAge() method is a getter that returns the age.
- ➢ The setAge() method is a setter that updates the age.
- ➢ The class Employee extends person shows inheritance.
- ➢ The Employee class inherits properties from the person class.
- ➢ The empid and salary are additional attributes of the Employee class.
- ➢ The super() method is used to call the parent class constructor.
- ➢ The display() method is overridden in the Employee class.
- ➢ The super.display() calls the parent class display method.
- ➢ The Employee display method also prints employee ID and salary.
- ➢ The new Employee() is used to create an object of the Employee class.
- ➢ The emp1.display() calls the method to show all details.

SOURCE CODE:

```
class person
{     public name:string;
  private age:number;
  public  city:string;
constructor(n:string,a:number,c:string)
  {
    this.name=n;
    this.age=a;
    this.city=c;
  }
  display():void{
    console.log("Name:",this.name);
    console.log("Age: ",this.age);
    console.log("City: ",this.city);
  }
  getAge():number
 {
   return this.age;
 }
  setAge(a:number):void
  {
    this.age=a;
  }
}
class Employee extends person
{
   public empid:number;
   public salary:number;
constructor(n:string,a:number,c:string,i:number,s:number)
   {
    super(n,a,c);
    this.empid=i;
    this.salary=s;
}
   display():void
   {
    super.display();
    console.log("ID: ",this.empid);
    console.log("Salary: ",this.salary);
   }}
var emp1=new Employee("Pravin",16,"cbe",3,1000);
emp1.display()
```

OUTPUT:

```
PS C:\Users\pravi> cd "E:\HTML"
PS E:\HTML> npx tsc oops.ts
PS E:\HTML> node oops.ts
Name: Pravin
Age:  16
City:  cbe
ID:  3
Salary:  1000
```

RESULT:

A TypeScript application is created to demonstrate object-oriented programming concepts like classes, inheritance, encapsulation, and method overriding.